🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements within the Same System

Lesson 3-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to convert between units within the same measurement system.
Language Objective: I can explain a conversion using the words unit ratio, convert, measurement system, and per.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Convert Measurements within the Same System | Conversion factor | Unit ratio | Convert | Measurement system | Double number line | Ratio table

👆 *Tap any word to see what it means and a picture.*

1. Feet into inches stays inside the ________ measurement system.

2. The ratio that trades one unit for another is a ________ ________.

3. A ratio comparing an amount to exactly ________ unit is a unit ratio.

4. To change a measurement to another unit without changing the amount is to ________.

5. Inches and feet belong to the customary ______________ ______________ ; centimeters and meters to the metric one.

6. Two lines whose matching ticks show the same amount in two units form a ______________ ______________ ______________ .

7. A table whose every column holds the same comparison is a ______________ ______________ .

Write About the Math

How many cup-scoops of broth does the recipe need, and how do you know?

¿Cuántas tazas de caldo necesita la receta y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Convert Measurements within the Same System** — *Convertir medidas dentro del mismo sistema*
- ☐ **Conversion factor** — *Factor de conversión*
- ☐ **Unit ratio** — *Razón unitaria*
- ☐ **Convert** — *Convertir*
- ☐ **Measurement system** — *Sistema de medidas*

Model: *Find the scale factor (120 divided by 8 = 15), then multiply each ingredient by it.* — Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The unit ratio is ____ cups to 1 quart. Cups are ____ than quarts, so I ____ to convert. 3 × 4 = ____ cups. The amount of broth did not change — only the ____ did.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.

- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.
Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Convert Measurements within the Same System
2. **Notes 2:** Conversion factor
3. **Notes 3:** Unit ratio
4. **Notes 4:** Convert
5. **Notes 5:** Measurement system
6. **Notes 6:** Double number line
7. **Notes 7:** Ratio table
8. **Watch for this mistake:** *A common mistake when converting within a measurement system is choosing the operation by habit instead of by unit size — multiplying every time. Converting 36 inches to feet by multiplying gives 36 × 12 = 432 feet, taller than a 40-story building. Ask first which unit is smaller: it takes MORE small units to make the same amount, so going to a smaller unit multiplies, and going to a larger unit divides.*
9. **Try It 1:** *— Every unit ratio says how many of the smaller unit fit inside exactly 1 of the larger one.*
10. **Try It 2:** *— Conversions in this lesson stay inside one system — customary with customary, metric with metric.*